We thank all five reviewers for their careful reading and are encouraged that they find the work worthwhile: "sensible" (27Jx), "important"/"promising" (D8hx), and "a fair and interesting contribution" with "compelling" matched-setting results (wGgF). We take the presentation problems seriously and have completed a full language and formatting revision, verifying that no model, experiment, or reported result is affected by these edits.

Reviewer 27Jx (#1) – R1-1: Presentation. We apologize for the draft-stage errors and thank the reviewer for the close reading. We have made a line-by-line revision: the dangling `Figure ??` now resolves to Fig. 1 (Sec. 3.2), the flagged typos are fixed (`multi-granul`→`multi-granular`, `Constrion`→`Construction`, `Fame-level`→`Frame-level`), and Figs. 1–2 are re-rendered as vector graphics. These are surface issues that affect no model, experiment, or result, and are fully correctable.

R1-2: Distinction from Vgent. The difference is paradigmatic, along two axes. ***Representation:*** Vgent builds a single ***flat*** graph over local frame/segment relations; we use an LLM to build a ***three-level hierarchical*** scene graph (frame→segment→event) unified by a shared entity registry, exposing cross-scale structure a flat graph cannot. ***Reasoning:*** Vgent is a fixed retrieve-then-read RAG pipeline that answers in a single pass; our agent treats the hierarchical graph as its primary retrieval index and reasons ***agentically*** over multiple rounds, issuing multi-level graph queries, verifying each proposal against raw frames, and deciding its next step until it converges. Together these drive the +18.2-pp matched-setting gain on LVBench (54.8%→73.0%).

R1-3: The role of the event graph. We respectfully find the 64.6%→64.4% reading too narrow: it isolates the Stage-3 ***increment*** and asks it to carry the contribution alone. ***(1)*** Our representation is a three-level stack (Base (frame-level) + Stage 2 (segment-level) + Stage 3 (event/video-level)), so the fair comparison is the full stack against Base, where the hierarchy improves ***both*** axes: accuracy rises 54.8%→73.0% (+18.2 pp, matched) and the event layer cuts LLM and frame-inspection calls by 71.5% and 74.7% and shortens the agent's reasoning rounds. ***(2)*** In our ***follow-up experiment*** that strengthens Stage 3, the event layer also lifts accuracy whilekeeping the efficiency: it reaches ***68.5%*** e2e on LVBench, +3.9 pp over the segment-only version (64.6%) and +13.7 pp over the caption-only Base (54.8%). We will report these strengthened, matched results in the revision.

Reviewer D8hx (#2) – R2-1: Offline construction cost. For all 103 LVBench videos (115.5h), construction costs $112.74 (12,360 `gpt-4.1` frame-graph calls at one keyframe per 5s, ≤120/video, 7.94h on four workers, and the symbolic Stage 3 event build adds only 121s), i.e. $0.98 per video-hour. We reuse DVD's open-sourced captions rather than generating our own, so this is our only added offline cost. End-to-end VQA over all 1549 questions costs $279.54 (`o4-mini`) and $499.31 (`o3`), so this one-time construction is only 40% and 23% of the recurring online budget it serves.

R2-2: Cross-benchmark control. Thanks for pointing this out. On LVBench the DVD comparison is fully controlled: the same `o3` backbone, DVD's own open-sourced captions, and the same video database. On the other three, DVD released no full reproduction setup, so we cannot match its pipeline. We still use its captions and report these as best-effort positions. Other commercial and open models have closed weights or fixed pipelines, so they are reference positions only. Our claims therefore rest on the fully controlled LVBench comparison and its matched-setting ablations.

R2-3: Direct graph quality. We agree direct graph-quality evaluation is valuable, but there is not yet a widely accepted hour-scale benchmark with open-vocabulary entities, relations, temporal alignment, and long-video annotations. We therefore follow prior long-video agent work such as DVD: caption/graph indices are evaluated by their end-to-end QA effect rather than by an intrinsic "agent-index quality" score. To make this more concrete, we also manually checked a small sample of generated graphs; it shows 76.8% entity-mention absorption and only a 5.0% invalid-relation rate. In the revision, we will report this human-verified sample together with the caption-free graph-only result, where the answer must come from the graph.

R2-4: Implementation details. To give the values asked for, entity alignment scores name, description, and type similarity with weights 0.7, 0.2, and 0.1, and merges above a threshold of 0.78. The entity blacklist removes artefacts and over-generic terms, and segments merge into events at a 12s temporal gap with relation-overlap thresholds. The composite retrieval score and all remaining thresholds are fully specified in the appendix, which gives the alignment, gating, and event-construction algorithms. We will release a fully reproducible repository.

R2-5: Presentation. Thanks. Please see R1-1.

R2-6: Isolating localization from FTT. Our reported answered accuracy (ans_acc) separates the two: the caption-only baseline scores 65.7% and SGDAgent 74.2% (+8.5 pp), independent of the answer rate, while the FTT drop (16.5%→1.6%) is itself a consequence of structured retrieval converging where open caption search loops for lack of locatable evidence.

Reviewer k56s (#3) – R3-1: Baseline data citation. Thank you for pointing out this. The correct DVD values are Video-MME 67.3 (we had mis-copied DVD's LongVideoBench(Long) entry, 68.6) and EgoSchema 76.6, both matching DVD's published numbers. The correction in fact widens our Video-MME margin to 69.0 vs. 67.3. We will present a carefully re-verified version, with a more general Fig. 1 example, in the revision.

R3-2: Related works. We will cite and discuss all three ([1] Mind Palace, [2] the Gaussian-token work, [3] GraphVideoAgent) in the revision. They use the graph as an in-context representation or online entity memory, whereas we use it as the agent's primary retrieval index, gated to raw-frame verification before answering (Please see R1-2), a distinct design.

R3-3: Offline construction cost. Thanks. Please see R2-1.

R3-4: Error propagation & robustness. Two mechanisms limit it. First, the propose-and-verify step gates every graph hit to raw-frame re-inspection before answering (R2-3), so a hallucinated or mis-aligned node is caught at verification rather than read out as an answer, reducing upward propagation. Second, the caption-free graph-only system still reaches 68.5% (Please see R1-3), evidence that the pipeline tolerates imperfect frame-level extraction. To quantify this we will add a robustness analysis that perturbs frame-graph quality and reports the accuracy sensitivity, and we will discuss [4].

Reviewer SSkz (#4) – R4-1: Over-strong coverage claim. We agree, and will revise the wording. The method improves ***localization within the sampled frames***, and we will no longer claim to address incomplete sampling coverage, since the graph itself is built only from sparsely sampled keyframes.

R4-2: The role of the event graph. In the submitted table the event graph is not a fixed-backbone accuracy driver, and we present it as a higher-order layer that reduces the agent's search burden. A strengthened, caption-free run, however, shows it also carries accuracy, not only efficiency (Please see R1-3).

R4-3: Heuristic construction. Please see R2-3 for robustness. We now fully specify the alignment, merging, and event thresholds (Please see R2-4).

R4-4: Presentation. Please see R1-1. We have already re-rendered Figures.

R4-5: More questions. Please see R2-3 for scene-graph quality evaluation. Repositioning the event graph as an efficiency layer is appropriate because it reduces search burden, but a strengthened caption-free run shows it is also accuracy-bearing (Please see R1-3). Please see R2-1 for full pipeline cost per video. For the sampling interval, we currently sample one frame every 5s and will report 2s/5s/10s sensitivity in the revision.

Reviewer wGgF (#5) – R5-1: HIG & hierarchy novelty. Thank you. We agree HIG proposes a hierarchical video graph, and we will cite and contrast it. We will not claim the hierarchy concept as novel. Our work differs in task: HIG targets video scene-graph ***generation***, whereas we use a graph index to ***drive an adaptive agent*** for long-video QA (Please see R1-2).

R5-2: Implementation details. The alignment, retrieval, gating, controller-signal, and event-construction rules are all specified operationally in the appendix as pseudocode. Please see R2-4 for the headline values.

R5-3: Graph-depth ablation. Please see R1-3. The segment graph drives fixed-backbone accuracy, and the event graph reduces search burden and, under a stronger construction, also contributes accuracy (caption-free 68.5%).

R5-4: Efficiency evidence. Beyond online call counts we now also report dollar cost and construction time (Please see R2-1).

R5-5: Presentation. Thanks. Please see R1-1.